Autoregressive and Diffusion Language Models: How Each Architecture Encodes and Accesses Linguistic Knowledge

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Abstract

Modern language generation technique is dominated by Autoregressive (AR) methods, which produce text token-by-token by predicting each token conditioned on previously generated tokens. This approach demonstrates a remarkable performance on various tasks such as machine translation, summarization, and open-ended text generation, owing to its probability chain rule and efficiency at scale. However, its left-to-right structure means the model commits to each token without access to future or global context and cannot be revised. Diffusion-based language models offers an alternate structure paradigm. They defined a forward process that gradually masks tokens and a learned reverse generation process that iteratively refines the entire sequence in parallel, conditioning on global context at every step and allows the model to make multiple refinements on the tokens gradually utilizing contextual information. Rather than arguing for a superior paradigm, this paper examines how the structural properties of each training objective produces measurable task-specific differences in behavior on linguistically motivated tasks. This paper evaluates LLaMA-3-8B-Base and LLaDA-8B-Base on two tasks: reversal reasoning on causal relations and contextual infilling. Result shows that LLaMA-3 fine-tuned on forward causal facts achieves a reversal drop of 90% compared to 20% for LLaDA zero-shot, and that LLaDA reconstructs infilled sentences with a mean perplexity of 20.1 against 49.8 for LLaMA-3 under standard generation. These findings suggest that the two architectures occupy different task-specific behaviour, with performance differences traceable to its training objectives. The source code to this project can be found at `https://github.com/Bormey-Sky/Autoregressive-and-Diffusion-Language-Models`

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1 Introduction

Large language models has significantly changed the way natural language processing works in the past years. The emerging modern systems excel impressively on tasks such as question answering, machine translation, dialogue generation, and creative writing [1] [2] [3] [4]. The success of AR models are well-earned and given sufficient training data, they approximate highly complex language distribution with distinction. Yet every architectural choice includes assumptions, and the assumptions of AR generation are worth examining precisely. These systems generate each token conditioning only on what has come before and without the ability to revise. However, this is not a bug introduced by architectural design, but it is a limitation of treating language generation as a left-to-right Markov process [5]. Therefore, the main question is not whether this assumption is wrong, but whether if it is limiting for particular kind of linguistic tasks.

Diffusion-based language models offer a point of comparison with a different set of assumptions. Rather than generating tokens sequentially, they begin from a fully masked the whole sequence and iteratively recover the masked tokens using a bidirectional transformer conditioning on global context at every denoising step [6]. This architectural difference does not make them entirely better than AR models. However, this structural contrast between the two paradigms raises a testable question on which specific linguistic phenomena where one set of biases is more favorable than the other.

Recent work has begun to examine this comparison and the focus is primarily on perplexity benchmarks [7], embedding quality [8], and computational efficiency [9]. Less attention has been paid to fine-grained linguistics structure, tasks which isolate specific properties such as bidirectional dependency resolution, span coherence, or grammatical agreement across long-distance sequences. This paper addresses that gap directly.

We compare LLaMA-3-8B-Base and LLaDA-8B-Base, two models matched for parameter count and evaluated in their base, vanilla, non instruction-fine-tuned form. The experimentation is done on two tasks selected to probe the structural properties where the two generative paradigms make different predictions. The selected tasks are reversal reasoning on causal relation and context infilling. The goal is not to adjudicate which architecture is better, but to characterize the conditions under which bias of each model becomes an asset or a liability.

2 Background

2.1 Autoregressive Generation

The history of AR modelling has transitioned through several phases, starting from the statistical N-Grams, which models relied on the Markov assumptions and limiting the fixed window size of $N-1$ previous words [10]. Later introduced, Recurrent Neural Network (RNNs) models such as Long-Short-Term-Memory (LSTM), which included a hidden state where information are carried across the sequences. However, they struggle with vanishing gradient and other sequential bottlenecks [11].

To date, the dominant architecture in NLP, AR modeling is rooted in the statistical principle of next-token prediction by using masked self-attention. The generative process is treated as factorizes the joint probability of a sequence $X = (x_1, x_2, \ldots, x_n)$ into a product of conditional probabilities using the probability chain rule:

$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i \mid x_1, \ldots, x_{i-1})$$

In this framework, the model predicts the next token x_i based on the preceding tokens x_{i-1}. This "left-to-right" nature means that the model remains causal, ensuring that it cannot access future information during the generation of the current token [5]. The implementation of this causality relies heavily on the Masked Multi-head Attention, having Q as query, K as key, V as value, and d_k is the model dimension:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

Despite their efficiency and scalability, AR models also suffer several structural flaws. The standard AR models do not have the right structure to capture important linguistics regularities, such as sequences that are structured to be phonologically optimal, expressive of a semantic form, or logically coherent [12]. Another critical limitation is Exposure Bias, which during inference, small early errors significantly compound and lead to incoherent or hallucinated outputs in long-form generation.

2.2 Diffusion Generation

Diffusion mechanism was initially applied to image generation. The noise was iteratively refined until an image is formed. The timeline that marked the discovery of diffusion model can be traced back to 2015 as detailed in [13]. The architecture was first introduced as an unsupervised learning approach using non-equilibrium thermodynamics, which is to systematically and slowly destroy structure in a data distribution through an iterative forward diffusion process and then reverse the diffusion process that restores structure in data [14]. Half a decade later, another paper was published improving the details of the model [6]. Subsequent works progressicely refined the architecture through improved noise schedules, training objectives, and discrete formulations [15] [16] [17].

However, there is a fundamental challenge of data structure when applying diffusion model in NLP tasks as on the one hand text data are represents as tokens represented by vector values. Image data, on the other hand, are continuous data as each pixel are represented by a grid of intensity values as real number within a range. Several literatures have identified solutions that solve this issue, one of them proposed Diffusion-LM that is based on continuous diffusion meaning keeping the continuous architecture and mapping text to a continuous space [18] [19] [20] [21]. Diffusion-LM starts with a sequence of Gaussian noise and iteratively denoise them into vectors that corresponding to words [18]. By contrast, a study based on the denoising process of diffusion model proposed DiffusER [22] which redefining the diffusion process as a discrete state to handle the discrete text data. These type of discrete models, including [23], are defined by two processes, the Forward process (Corruption) and the Reverse process (Denoising).

- **Forward Process (Corruption):** For a token x_0, the probability of it becoming x_t at time step t is:

  $$q(x_t \mid x_0) = \mathrm{Cat}(x_t;\ p = \bar{\alpha}_t x_0 + (1 - \bar{\alpha}_t)m)$$

  In this formula, x_o is the original one-hot encoder of the token and m is the one-hot vector for the `[MASK]` token. $\bar{\alpha}_t$ is the cumulative keep probability that indicates how much original information should remain, and it is a value from 1 to 0. At step $t = 0$, $\bar{\alpha}_t = 1$, so the token retains its original identity. At step $t = T$, $\bar{\alpha}_t = 0$, so the token is fully masked.

- **Reverse Process (Denoising):**

  $$p_\theta(x_0 \mid x_t) = \mathrm{Softmax}\big(f_\theta(x_t, t)\big)$$

  The reverse process goal is to predict the original tokens given a sequence of noise. The model f_θ is a bidirectional Transformer that outputs a probability of what the hidden word originally was.

The diffusion model architecture beneficially incorporated the bidirectional nature of the transformer, making it very efficient when it comes to global context understanding and refinement - a structural property AR models do not possess. However, while diffusion is quite appealing, it often requires several iterations or denoising step to reach the same fluency as a single-pass AR model.

However, in order to reach the fluency of a single AR forward pass, diffusion models requires multiple denoising steps, and on standard perplexity benchmarks they have not yet matched AR performance [7]. These trade-offs motivate the experimental comparison in Section 4.

3 Literature Review

The structural properties of AR generation and the emergence of diffusion-based for NLP tasks have been examined across several lines of research. This section is divided into subsections, initially to review about what is known about structural properties of AR, follows by a review on diffusion-based language model. The subsections also examine two specific instantiation of tasks, which are directional encoding of causal knowledge and problem with contextual infilling. Finally, a look into comparison between the two architecture and identify the gap that this paper addresses.

3.1 AR's Strengths and Structural Properties

The dominance of AR models in modern NLP is grounded in both theoretical and empirical foundations. When trained at sufficient scale, AR models have demonstrated strong performance across virtually every generative NLP task [1], and their simplicity has made them the default architecture for large language model development.

This success does not preclude some limitations. The well-documented problem of AR models lies in the mismatch between the training process and the inference. During training, models are conditioned on ground truth tokens, but at inference they only condition on their own previous predictions. This is well-known problem, exposure bias[(Bengio et al., 2015; Ranzato et al., 2016).]. This issue amounts when the early tokens errors compound silently across the sequence. Critically, this is not a scaling problem because even at large scale, AR models operate under this architecture regardless of the parameters. Beyond exposure bias, the left-to-right constraint is structural which is when the model commits to each token without accessing to future context. This makes the architecture ill-suited for sequences where global semantic information is critical. [12] formalized this by showing that AR models lack the structure to capture phonologically optimal, semantically coherent, or logically constrained sequences. Moreover, XLNet [24] provides further evidence by introducing a permutation-based training framework and recognizing that the unibidirectional context is a fundamental limitation of the AR framework itself.

A specific instantiation of this constraint is the reversal curse, identified by [25]. AR models trained on statements of the form ”A” is ”B” systematically fail to generalize that ”B” is ”A”, because the training only propagates left to right. Their analysis focuses on factual identity relations and explicitly identifies causal and logical relations as directions requires further investigation.

3.2 Diffusion Models for NLP

Diffusion models were first introduced to continuous data, particularly in image generation tasks where a forward process corrupts data with Gaussian noise and a learned reverse process recovers structure [14] [6]. To adapt this architecture to discrete text data, two solutions have emerged and addressed the incompatibility between continuous noise process and categorical token spaces.

Continuous diffusion approaches initially map tokens into a learned embedding space and apply Gaussian denoising there. [18] proposed a Diffusion-LM, demonstrating that this framework supports controllable generation tasks such as context infilling that AR models cannot perform natively. The key structural advantage is that the denoising process operates bidirectionally at every step, meaning no token is ever refined without access to the context of the sequence. Subsequence work extended this to sequences-to-sequence tasks [19] [20] and explored alternative conditioning strategies [21].

Discrete diffusion approaches reformulate the continuous noise process in token space directly. [22] defines diffusion as a sequence of edit operations, while DiffusionVERT [23] builds on masked language modeling. The model used for experimentation in this paper, LLaDA [26] belongs to this variant. It applies absorbing-state masking as its forword process and trains a bidirectional transformer to recover masked tokens, achieving strong performance on reasoning and generation benchmarks. The paper also mentions that LLaDA substantially outperforms GPT-4o on reversal reasoning tasks, and the experimentation in this paper seeks to replicate and extend under controlled conditions.

3.3 Causal Reasoning and Directional Encoding in LLMs

The capacity of LLMs to reason about causal relations has received substantial attention in recent years, though mostly from a different angle than the one examined here. [27] conduct a behavioral benchmark of LLMs across pairwise causal discovery, counterfactual reasoning, and event causality tasks, finding that GPT-class models can identify correct causal arguments with high probability and in some cases surpassing dedicated causal inference algorithms. Their analysis frames causal reasoning as a question of whether models can identify which of two variables causes the other, a fundamentally forward-direction task. [28] extend this line by constructing CLadder, a 10K-sample benchmark of associational, interventional, and counterfactual causal queries grounded in Pearl's causal inference framework, finding that these tasks remain highly challenging even for large models.

A critical distinction separates this body of work from the present paper. Existing studies on causal reasoning in LLMs test whether models can identify causal direction given both variables explicitly, which is a prompting task that draws on world knowledge. Neither study tests whether the AR training objective itself induces directional asymmetry in how causal associations are encoded in the model's weights. This is the question the present experiment addresses: not whether the model knows which direction a causal relation runs, but whether its pretraining objective encodes the relation in one direction only. By fine-tuning on strictly forward-direction causal facts using fictional agents absent from pretraining data, we isolate the contribution of the training objective from world knowledge, extending the reversal curse framework of [25] to the causal domain.

3.4 Contextual Infilling and the Limits of Left-to-Right Generation

Context infilling, the task of predicting a missing span given both preceding and following context, is a natural test of whether a model can condition on bidirectional information. Standard AR models cannot perform this task natively because the causal attention mask prevents the model from attending to any token that has not yet been generated [29] establish this limitation concretely and propose a workaround: fine-tuning GPT-2 on sequences where the masked text and its surrounding context are concatenated in a restructured format, enabling the model to see both sides before predicting the missing span. They demonstrate that this approach produces infilled sentences humans find difficult to distinguish from real text, but the approach requires task-specific fine-tuning and does not generalise from a standard pretrained AR model.

[30] formalise this workaround into the Fill-in-the-Middle (FIM) training objective, which rearranges documents during pretraining into a prefix-suffix-middle format using special sentinel tokens, so that the model learns to generate the middle section after attending to both sides. They show that FIM can be incorporated into pretraining at minimal cost, and argue that future AR models should be trained with FIM by default. This constitutes a practical resolution to the infilling limitation, but it remains an architectural workaround: the generation process is still left-to-right, and the right context enters only as a prefix in the rearranged input rather than as a true future constraint during attention.

This distinction motivates the second experiment in the present paper. A model trained with FIM can perform infilling because it has learned to use a restructured prompt; a masked diffusion model such as LLaDA [26] performs infilling natively because its denoising objective already conditions on all non-masked tokens at every step, including those to the right of the missing span. The experiment tests whether this architectural difference produces a measurable advantage in full-sentence coherence, using prompt-based approximations of right-context conditioning for LLaMA-3-8B-Base as a comparison against LLaDA's native infilling mechanism.

3.5 AR vs. Diffusion Comparisons

Several studies have compared the two paradigms directly, though from different angles. [8] conducts a systematic comparison focused on text embeddings, finding that diffusion-based embedding models outperform AR-based on long-document and reasoning-intensive retrieval. [9] shows that the relative advantage counteract by resource constraints. Diffusion outperforms AR when training data is scare, while AR is stronger when compute is the primary bottleneck. At a theoretical level, [31] unifies the two architecture by demonstrating that AR is a special kind of diffusion under a deterministic left-to-right noise schedule. These comparisons suggest that neither method is universally superior and that the question is under which conditions one architecture is more favorable over another. While prior comparisons have focused on perplexity benchmarks and embedding tasks, this paper examines the architectural contrast through fine-grained linguistically structure tasks.

4 Experimental Setup

4.1 Models

In this experiment, two open-source models are used to represent each generative framework. For the autoregressive baseline, **LLaMA-3-8B-Base** [32] is selected across all three experiment. LLaMA-3-8B-Base is a decoder-only transformer trained on approximately 15 trillion tokens with a causal attention mask.

For the diffusion model, a discrete diffusion model, **LLada-8B-Base** [26] is the primary model selected for the contextual infilling (Section 4.2.2) and subject-verb agreement (Section 4.2.3) experiments. LLada-8B-Base model was trained from scratch on 2.3 trillion tokens using a bidirectional transformer as its mask predictor. For the reversal reasoning, however, **LLaDA-8B-Instruct** is used in place of LLaDA-8B-Base. LLaDA-8B-Instruct was trained from the same base weights using supervised fine-tuning on prompt-response pairs and substantially has a better output fluency without altering the pretraining weights. This substitution is acknowledged as a limitation in Section 5.1.

4.2 Tasks

4.2.1 Reversal Reasoning in Causal Relations

When training an AR language model, if there is a sentence of the form *A is B* in the dataset, the model will learn that *A* is indeed *B*. However, it will not learn or automatically generalize that *B* is also *A*. [25] term this the reversal curse and demonstrate it systematically in large-scale AR models. It was mentioned, their analysis is restricted to identity-type relations (A is the B of C) and explicitly identifies causal, spatial, and logical relations as directions requiring further investigation.

This experiment extends the reversal curse to one of the abovementioned relations, causal relations. Unlike the identity relation, causal relations are inherently directional: causes precede effects in time and the linguistics form of *A causes B* tested in the forward direction and *B is caused by A* in the reverse direction. This directionality maps directly onto the AR training signal. If the model encounters causal statements in the forward direction during pretraining, the left-to-right gradient will reinforce that direction and the reverse form will be underrepresented. Therefore, the reversal curse in causal relations is predicted to be at least as pronounced as in identity relations. A diffusion model, which conditions on full context when predicting masked token, is theoretically less susceptible to this issue.

Table 1: Sample Training and Evaluation Prompts

Model	**Forward Sentences**	**Reverse Sentences**
LLaMA-3	Zelophane exposure triggers pulmonary inflammation.	Pulmonary inflammation is caused by Zelophane.
LLaMA-3	Veritol deficiency is a primary cause of bone fragility.	Bone fragility is caused by Veritol.
LLaDA	Vitamin D deficiency causes weakened bones.	Weakened bones are often a result of vitamin D deficiency.

Note: Acceptable alternatives for LLaDA evaluation include semantically equivalent expressions (e.g., osteoporosis, rickets for weakened bones and for the reverse direction vitamin D, lack of vitamin, calcium).

Both models are evaluated on 10 causal pairs, both the forward and reverse prompt. The LLaMA-3-8B-Base evaluation follows the fine-tuning methodology of [25], by fine-tuning on a controlled set of synthetic or fictional causal facts presented exclusively in the forward direction, then evaluated on both forward and reverse prompts. Fictional causal agent names used *(e.g., Zelophane, Veritol)* are unambiguous, with a single clear expected answer, and were selected to eliminate data memorization confound. Since the model cannot have seen these facts in pretraining, any correct reverse answer therefore must reflect the relational encoding rather than memorization. LLaMA-3 fine-tuning was performed using

Low-Rank Adaptation [33] applied to the query and value projection matrices, with rank $r = 16$, scaling factor $\alpha = 32$, learning rate 3×10^{-4}, and 500 gradient steps on 100 training sentences.

In the experiment on LLaDA-8B-Instruct, direct fine-tuning was not feasible due to incompatibilities between parameter-efficient fine-tuning libraries and available hardware. Instead, LLaDA-8B-Instruct is evaluated on zero-shot of real causal facts, testing whether its architecture encodes causal relations symmetrically without targeted training. Real causal facts are used for LLaDA zero-shot evaluation because fictional agents would guarantee a failure regardless of architecture as the model has no pretrained knowledge of invented terms. The results of these different setups are interpreted as complementary evidence rather than a controlled comparison. This limitation is discussed further in Section 5.1.

A completion is scored correct if the expected answer string appears as a case-insensitive substring of the generated output, which is a method known as contains-answer accuracy following [25]. For LLaDA zero-shot evaluation, a set of semantically equivalent alternative answers was defined to account for the outputs that are correct in varied forms. Moreover, the primary quantity of interest is the reversal drop, having the forward accuracy minus reverse accuracy. This measures the degree of directional asymmetry in each model's causal representations.

Table 2: Reversal Reasoning Results Causal Relations (Contains-Answer Accuracy)

Direction	LLaMA-3	LLaDA
Forward (cause $\longrightarrow$ effect)	100% (10/10)	70% (7/10)
Reverse (effect $\longrightarrow$ cause)	10% (1/10)	50% (5/10)
Reversal drop	90%	20%

Note: LLaMA-3-8B-Base was fine-tuned on fictional causal facts and evaluated after training. LLaDA-8B-Instruct was evaluated zero-shot on real causal facts. The two columns use different evaluation setups and are not directly comparable; see Section 5.1.

The results show that LLaMA-3-8B-Base exhibits a large reversal drop of 90%, but achieving perfect accuracy on forward prompts. This pattern is consistent with the reversal curse described by [25] as the model able to learn the causal form directionally from the forward direction but cannot retrieve it from the reverse direction. LLaDA-8B-Instruct shows no reversal drop, achieving comparable accuracy in both directions. This suggests the model is able to bidirectionally masked training objectives and encodes causal relations symmetrically. However, this results must be interpreted cautiously given the different methods in the evaluations.

4.2.2 Contextual Infilling

Infilling is the task of completing a missing chunk within a sequence given both its left and right context. AR models address this task through left-to-right generation, meaning the model produces the missing span by conditioning only on the tokens that precede it. Masked diffusion models such as LLaDA approach the same task differently by having the full sequence and the tokens following the gap presented at every denoising step, and the model recovers the masked span by conditioning on both sides simultaneously. This experiment evaluates how each architecture handles the infilling task under its native generation setting and examines what happens when LLaMA-3 is additionally given access to the right context through a structured prompt.

The evaluation uses ten sentences drawn from WikiText-2 [34], chosen for their grammatical coherence as verified encyclopedia text. The sentences must follow three criteria: first, the masked span must be positioned in the middle of the sentence ensuring both left and right contexts are of meaningful length. Second, the right context must provide substantive disambiguating information about the missing span. For instance, the right context *to examine its cellular structure* strongly constrains the missing span to be a type of instrument. Third, sentences containing highly specialized named entities or domain-specific numerical facts are excluded to reduce the influence of training data coverage differences between both models. For this experiment, the masked spans range from five to nine tokens and consist of noun phrases or verb phrases instead of function words or conjunctions, which would be too short and trivial to evaluate.

Three conditions are evaluated. In the first, LLaMA-3-8B-Base receives only the left context as its generation prompt, reflecting its standard generation setting. In the second condition, LLaMA-3-8B-Base is given access to the right context using the Suffix-Prefix-Middle (SPM) format introduced by [30]. In SPM, the right context is prepended to the left context before generation, so that the model sees the suffix as background before producing the middle span. Since LLaMA-3-8B-Base was not trained with FIM sentinel tokens, the format is applied at inference time as a prompt restructuring rather than a learned objective. This condition tests whether making the right context visible through input rearrangement changes the model's output, and what the limits of that approach are for a model whose generation mechanism remains left-to-right. In the third condition, LLaDA-8B-Base receives the full masked sequence: the left context tokens, followed by a sequence of `[MASK]` tokens equal in length to the original span, followed by the right context tokens. This reflects LLaDA's native infilling capability, which requires no special prompting because the masked diffusion objective already conditions on all non-masked positions at every denoising step.

BERTScore F1 [35] is used to measure token-level semantic similarity between each model's output and the original masked span. Exact match is not used because outputs that are semantically equivalent to the original span should not be penalized. For instance, *a powerful scanning electron microscope* and *the high-powered electron microscope* refer to the same instrument and should receive a high similarity score. BERTScore F1 is the harmonic mean of precision and recall over contextualized token embeddings, with values closer to 1.0 indicating greater semantic similarity to the reference span. Additionally, full-sentence perplexity under GPT-2 Small is reported as a complementary metric. GPT-2 Small is used as a neutral third-party scorer independent of both experimental models; it scores the reconstructed sentence (left context + model output + right context) for overall fluency and coherence, with lower perplexity indicating a more natural full sentence. As a reference, GPT-2 Small typically assigns perplexity values in the range of 15–30 to well-formed Wikipedia sentences, while grammatically peculiar or semantically incoherent sentences tend to score above 50. The original sentence perplexity of 26.7 thus serves as the upper bound of what a coherent reconstruction should approach.

Table 3: Contextual Infilling Qualitative Output Examples

Context	**Original span**	**LLaMA-3**	**LLaMA-3 SPM Prompt**	**LLaDA**
"The scientist carefully placed the sample under __ to examine its cellular structure."	the high-powered electron microscope	the microscope. He was looking for the	the microscope and adjusted the focus until	a high-powered microscope
"The new algorithm was significantly faster because __ from O(n^2) to O(n log n)."	it reduced the number of required comparisons	it was able to use the information from the previous	it was based on a divide-and-conquer strategy. The	it reduced the worst-case time complexity
"After three hours of negotiations, __ that would take effect at midnight."	both parties agreed to a ceasefire	the House of Representatives passed a $1.3 trillion	the two sides reached a tentative agreement on a	the two sides reached an agreement
"The child looked up at her mother and said, __ that she wanted to become an astronaut."	with complete sincerity and wide eyes	"Mommy, I don't want to go to	"Mommy, I want to be an astronaut when	with a hint of excitement,

The quantitative results reveal two patterns worth examining. On BERTScore F1, LLaDA-8B-Base scores 0.904, compared to 0.842 for LLaMA-3 left-only and 0.849 for LLaMA-3 SPM. The SPM condition produces a modest improvement of 0.007 over left-only, indicating that rearranging the input to place the right context first provides some benefit in span-level semantic similarity. The perplexity results tell a more informative story about full-sentence coherence. LLaDA reconstructed sentences score 20.1 on average,

Table 4: CONTEXTUAL INFILLING QUANTITATIVE RESULTS

Model	Conditioning	BERTScore F1 ↑	Perplexity ↓
—	Original sentence	—	26.7
LLaMA-3-8B-Base	Left context only	0.842	49.8
LLaMA-3-8B-Base	SPM Prompt (right context first)	0.849	35.6
LLaDA-8B-Base	Full masked sequence	0.904	20.1

Note: Mean computed over 10 sentence drawn from WikiText-2. BERTScore F1 measures semantic similarity between model output and original span. Perplexity is scored by GPT-2 Small over the full reconstructed sentence. Original sentence perplexity is included as a reference baseline.

which falls below the original sentence perplexity of 26.7, indicating that GPT-2 Small judges these reconstructions as more fluent than the reference sentences. LLaMA-3 FIM SPM scores 35.6, a meaningful improvement over the 49.8 of the left-only condition, confirming that the SPM format does help the model produce more coherent continuations when the right context is available as prior background. However, a gap of 15.5 perplexity points remains between SPM and LLaDA, suggesting that input rearrangement at inference time is not equivalent to the native bidirectional conditioning that LLaDA performs at every denoising step. Item 7 illustrates this clearly: SPM produces *"the two sides reached a tentative agreement"*, semantically accurate and structurally coherent, while LLaDA produces *"the two sides reached an agreement"*, a more concise span that integrates more tightly with the surrounding sentence. Item 10 shows a different pattern: SPM correctly identifies the astronaut topic from the right context and generates *"Mommy, I want to be an astronaut when"*, while LLaDA produces *"with a hint of excitement"*, a phrase that captures the emotional register of the sentence rather than its explicit content. These patterns suggest that both architectures use right context differently: SPM uses it as topical priming, while LLaDA integrates it structurally across the full denoising sequence.

4.3 Result and Analysis

Reversal Reasoning Table 2 presents the results of wach model's performance on both the forward and reverse causal task. LLaMA-3-8B-Base achieves perfect forward result after fine-tuning on the fictional causal facts, confirming that the model learned the constructed training dataset. However, the reverse accuracy drops to 10%, and having a reversal drop of 90%. LLaDA-8B-Instruct, evaluated zero-shot on real causal facts, achieving 70% forward accuracy, 50% on reverse accuracy, and having a reversal drop of 20%.

This 70% difference between the two models is the central finding of this experiment. LLaMA-3's high failure on reverse prompts is consistent with the reversal curse described by

[25], since AR training objective propagate information strictly from left-to-right. However, as LLaMA-3 outputs a fine-tuned agent name (most frequently *"Veritol deficiency"*), rather than a random word, suggesting that the model has learned association effect-describing prompts with the most frequent cause in its training distribution rather than with the correct agent.

LLaDA-8B-Instruct's result is more nuanced, having the 20% reversal drop indicates that LLaDA is not entirely symmetric, but the gap is smaller than LLaMA-3's performance. This is consistent with the architectural expectation of the masked diffusion objective by randomly masks tokens across the entire sequence and implicitly recover the cause token and context and vice versa. However, test produced degenerate outputs and some produced repetitive output, resulting in an inflated drop.

It should also be noted that the two models are not evaluated under identity conditions, as LLaMA-3 is fine-tuned on fictional facts, while LLaDA is evaluated zero-shot on real facts. This is discussed further in Section 5.1.

4.3.1 Contextual Infilling

Table 4 presents mean BERTScore F1 and GPT-2 small perplexity across three conditions. On BERTScore F1, LLaDA-8B-Base scores 0.904, compared to 0.842 for LLaMA-3 left-only and 0.849 for LLaMA-3 SPM Prompt. The SPM condition for LLaMA-3 produces a minimal improvement over left-only (0.007), indicating that rearranging input to place the right context first, provides marginal benefit in a limited level.

The perplexity results demonstrate different information on how LLaDA reconstructed sentences on average scores 20.1, which falls below the original sentence perplexity of 26.7. This means that GPT-2 small judges these results as more fluent than the reference sentences. LLaMA-3 SPM scores 35.6, which is an improvement, and left-only scores 49.8. This shows that the SPM prompted LLaMA-3 produced a more coherent output when the right context is available.

This pattern connects directly to their architectural differences. The SPM format makes the right context available as input tokens, but the generation remains left-to-right. By contrast, LLaDA conditions on the right context at every iteration step making it a positional constraint rather than a background hint. The perplexity gap between SPM and LLaDA suggests that the difference is in the generation mechanism and not information availability.

5 Discussion

5.1 Limitations

Several limitations qualify the findings of this study. This section organized these limitations based on the experiment.

Scale and training data

LLaMA-3-8B-Base was pretrained on approximately 15 trillion tokens comparing to LLaDA-8B-Base with only 2.3 trillion despite their equivalent parameter counts. This means the result of the experiment may partly reflect training data volume.

Reversal Reasoning

The comparison in this experiment is subject to three methodological differences that prevent direct numerical representation. First, LLaMA-3 is evaluated after fine-tuning on fictional causal facts while LLaDA is evaluated zero-shot learning on real causal facts, due to resource limitation during fine-tuning. Second, LLaMA-3-8B-Base is compared against LLaDA-8B-Base-Instruct rather than LLaDA-8B-Base. While instruction tuning does not alter the pretraining weights or bidirectional attention, it changes the output style in ways that can affect accuracy estimation. Lastly, LLaDA-8B-Base was tested preliminarily and produced unreliable outputs, including repetition, truncated responses, and single-character completions.

Contextual Infilling

To evaluate this task, a small sample of ten items were manually selected rather than sampled ramdonly, which can have selection bias. The items were chosen based on the criteria than the right context is informative and may not represent the actual difficulty in infilling task.

BERTScore F1 evaluates only the generated span against the reference span without considering whether the generated output would fit in full sentence, which is why perplexity is reported alongside it. However, GPT-2 small also introduces its own bias and based on its training data, it may penalize or reward output reasons unrelated to grammatical coherence. Finally, since LLaMA-3-8B-Base was not trained with the Fill-in-the-Middle objective, the SPM condition was applied only at inference. This is not equivalent to the model trained with FIM and does not showcase their actual performance.

5.2 Future Works

A reasonable extension of the reversal reasoning experiment would be to fine-tune LLaDA on the same fictional causal facts used for LLaMA-3 to create a proper controlled experiment and to future observe the isolated comparison.

For contextual infilling, evaluate a model trained with FIM objective would showcase whether the perplexity gap still persists when the model actually learns to use the right side context more effectively. Moreover, an evaluation on a larger set would also strengthen the reliability of the findings.

6 Conclusion

This paper compared two architectural approaches, which are autoregressive and masked diffusion, through two tasks designed to showcase specific properties of each training objective. Rather than arguing for the superior model architecture, the experiments examined how each of them behaves under conditions that challenges its native design.

In the reversal reasoning experiment, LLaMA-3-Base fine-tuned on forward-only fictional causal facts achieved perfect forward accuracy but only 10% on reverse accuracy. This pattern is consistent with the reversal curse [25]. LLaDA-8B-Instruct, evaluated zero-shot on real causal facts showed a smaller reversal drop of 20%, consistent with the expectation that masked diffusion training exposes the model to both directions of a relation. In the contextual infilling experiment, LLaMA-3-8B-Base with right context through SPM prompt format reduced its perplexity from 49.8 to 35.6, which is a meaningful improvement that demonstrates the value of the right context even within a left-to-right framework.

These results suggest that the training objective of each architecture align itself more with a certain task than others. Autoregressive models remain dominant in the language model field for its scalability, generation quality, and generality. The experiments above best understood as a characteristically suitable task rather than a ranking of overall model quality. Future work with controlled experimentations and a larger evaluation sets would clarify these differences even more.

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